

Professional summary

Data Engineer at Capgemini

Data Engineer with 1.8+ years of experience building scalable data pipelines and cloud-based data platforms for healthcare analytics. Experienced in designing end-to-end ETL pipelines using AWS, Python, SQL, and PySpark to process large-scale datasets from multiple external systems.

Proficient in implementing event-driven data ingestion architectures, Medallion data lake design, and distributed data processing to support enterprise analytics platforms. Skilled in using AWS S3, AWS Glue, AWS Lambda, AWS Step Functions, and Amazon Redshift to build automated, reliable, and high-performance data pipelines.

Strong expertise in data transformation, data quality validation, deduplication using fuzzy matching, and performance optimization, enabling organizations to transform raw operational data into analytics ready datasets for business intelligence and reporting.

Experience

Data Engineer

August 2024 - Now
HCA Healthcare / United States, USA

TECHNICAL SKILLS:

Programming & Data Processing: Python, SQL, PySpark, Pandas, Apache

Sparfi Big Data Technologies: Hadoop, Hive, Databricks, Delta Lake

Cloud Platforms AWS: Amazon S3, AWS Glue, AWS Lambda, AWS Step Functions, Amazon Redshift, Amazon CloudWatch

Azure: Azure Data Factory, Azure Databricks, Azure Storage

Data Engineering: ETL Pipeline Design, Data Modeling, Data Quality Frameworks, Data Validation, Data Deduplication, Business Rule Processing, Large-Scale Data Processing

Databases & Tools: MySQL, SQL Server, Git, Jupyter Notebook, Visual Studio Code

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Education

Bachelor Of Commerce

2022 - 2025
Savitribai Phule Pune University
India, Pune

Skills

Python	◆◆◆◆◆
SQL	◆◆◆◆◆
Apache Spark	◆◆◆◆◆
Data Warehousing	◆◆◆◆◆
ETL	◆◆◆◆◆
Cloud Computing	◆◆◆◆◆
Data Modeling	◆◆◆◆◆
Big Data	◆◆◆◆◆
Data Analysis	◆◆◆◆◆
Business Intelligence	◆◆◆◆◆
Data Mining	◆◆◆◆◆
NoSQL	◆◆◆◆◆
Data Visualization	◆◆◆◆◆
Apache Hadoop	◆◆◆◆◆
AWS EC2 S3 Lambda Stepfunction	◆◆◆◆◆
RDS Redshift DynamoDB Aurora	◆◆◆◆◆

Project Overview

Pharma Insights: Enterprise Pharmaceutical Data Platform & Prescription Analytics

Pharma Insights is a **cloud-based pharmaceutical data platform built on AWS** designed to ingest, process, and analyze large volumes of healthcare partner data including **drug prescriptions, pharmacy sales transactions, hospital inventory data, and distributor shipment information**.

The platform integrates data from **multiple external healthcare partners and processes it through a modern data Lakehouse architecture based on the Medallion model**.

Using **AWS cloud services and distributed processing with PySpark**, the system automatically ingests **20–40 GB of daily data**, performs **data cleansing, standardization, deduplication, and transformation**, and produces analytics-ready datasets for **business intelligence dashboards and regulatory reporting**.

The platform significantly improved data reliability, reduced manual data preparation, and enabled **faster pharmaceutical sales analytics and operational insights**.

ROLES & RESPONSIBILITIES

Designed **event-driven data ingestion pipelines** using **Amazon S3, AWS Lambda, and AWS Step Functions** to automate the processing of partner data files.

Developed **AWS Glue PySpark jobs** to perform large-scale **data transformation, schema normalization, and data cleansing**.

Implemented **data validation frameworks** to verify **file structures, schema consistency, and required data attributes** before ingestion.

Built **Medallion data lake architecture (Bronze, Silver, Gold layers)** using **Amazon S3** to manage raw, processed, and analytics datasets.

Implemented **fuzzy matching algorithms using PySpark** to detect duplicate patient and prescription records using **similarity scoring techniques such as Levenshtein distance**.

Optimized large datasets using **partitioning strategies and columnar storage formats such as Parquet**, improving query performance and storage efficiency.

Designed **analytics-optimized schemas in Amazon Redshift** including **fact and dimension tables** to support pharmaceutical sales and prescription analytics.

Implemented **pipeline monitoring using Amazon CloudWatch** and configured **Amazon SNS alerts** for pipeline failures and operational events.

Worked closely with **data analysts and business teams to deliver high-quality datasets supporting sales analytics, prescription trend analysis, and compliance reporting**.

Athena | Glue | EMR | Kinesis



SNS | SES | SQS | ECR



SageMaker | SercetsManager | CloudWatch | EventBridge



Azure Blob Storage | Azure Functions | Logic App



Azure SQL Database | SQL Managed Instance | Synapse



Azure Data Factory | Resource Manager | ML Studio



Hobbies

- Traveling
- Reading
- Photography

Languages

- English
- Hindi
- Marathi
- Kannada